Applied AI & Robotics Portfolio

Academic Journey & Technical Capstones

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Academic Growth & Core Toolkit

- **Journey**: Foundational ML → Advanced Computer Vision → Enterprise AI Deployments
- **Environments**: Python, Google Colab, Jupyter, Git, Windows Command Line
- **Frameworks**: NumPy, Pandas, Scikit-Learn, OpenCV, TensorFlow, Keras, PyTorch

Core Course Capstones

- **Machine Learning**: Telco Customer Churn EDA Project
- **Computer Vision**: SentinelClean Intelligent Background Noise Removal Pipeline
- **AI Applications**: Real-Time Financial Fraud Anomaly Processing and Shielding Engine

Developed AI Capabilities

- **Data Pipelines**: Profiling distributions, cleaning messy spaces, feature engineering
- **Vision & Modeling**: Convolutional Neural Networks, YOLO object detection, matrix pixel transformations
- **Analytical Evaluation**: Confusion matrices, IoU scoring, precision-recall optimization, ethical bias mitigation

Horizons & Professional Contact

- **Reflection**: Mastered end-to-end data pipelines, complex model troubleshooting, and deployment strategies
- **Future Goals**: Scaling deep learning frameworks, autonomous robotics integration, edge AI deployments
- **Links**:

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